

Title: Ultrarapid hydroxylation of debrisoquine in a Swedish population

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Abstract: Hydroxylation of debrisoquine, catalyzed by the cytochrome P450 CYP2D6 exhibits genetic polymorphism, with large inter-individual differences in metabolic capacity. About 7% of Caucasians carry deficient CYP2D6 alleles and lack the CYP2D6 enzyme (poor metabolizers). We have shown in two Swedish families, individuals carrying duplicated or amplified functional CYP2D6L-genes (CYP2D6L2), causing the opposite phenomenon, ultrarapid metabolism of debrisoquine. In the present study, the occurrence of extra copies of CYP2D6L-alleles was studied in relation to debrisoquine metabolic ratio (MR) in 270 Swedish Caucasians including 64 selected subjects with very rapid metabolism ($MR \leq 0.2$). Thirteen of the 64 subjects carried a duplicated CYP2D6-gene as identified by EcoRI and XbaI restriction fragment length polymorphism and allele-specific polymerase chain reaction-amplification of genomic DNA. A new allele with three active CYP2D6L-genes was identified, characterized by an XbaI 54 kilobase fragment. This indicates a preference of the CYP2D6L-gene to be amplified compared to other CYP2D6 genes. Only one subject with an MR higher than 0.2 carried the duplicated CYP2D6L-allele, also being heterozygous for the defect CYP2D6B-allele. The overall frequency of the duplicated/amplified CYP2D6-allele was about 1%, and was present in 40% of subjects with MRs ≤ 0.1 . Thus, other variant CYP2D6-genes may exist that cause increased CYP2D6 activity. In conclusion, a haplotype with duplicated or amplified functional CYP2D6 genes predicts, with high accuracy, ultrarapid metabolism of debrisoquine. Genotyping for this CYP2D locus variant might be of value in patients not responding to generally recommended doses of CYP2D6 substrates, to distinguish between high metabolic capacity and noncompliance.